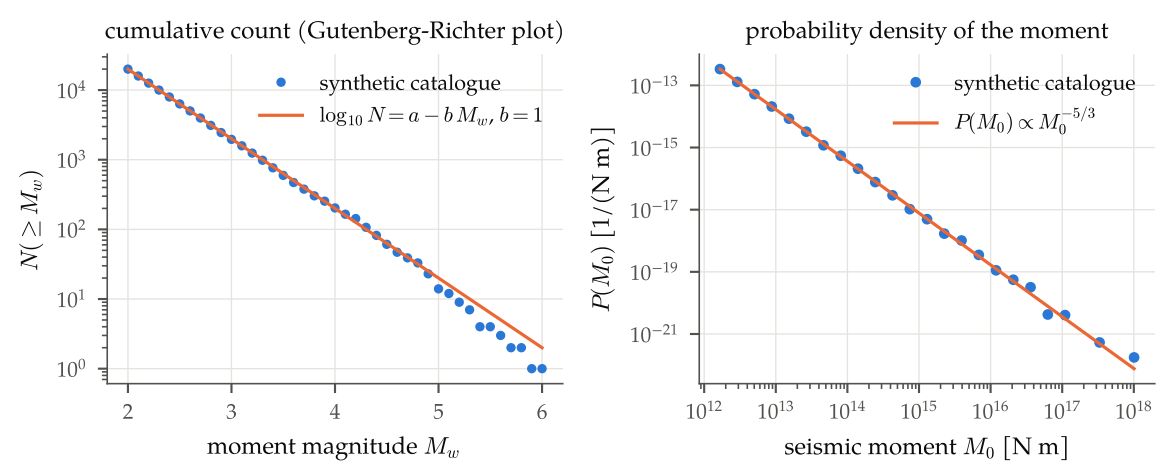
cumulative count (Gutenberg-Richter plot)
synthetic catalogue
$\log_{10} N = a - b\,M_w$, $b = 1$
$N(\geq M_w)$
10^4
10^3
10^2
10^1
10^0
2
3
4
5
6
moment magnitude M_w
probability density of the moment
synthetic catalogue
$P(M_0) \propto M_0^{-5/3}$
$P(M_0)$ [1/(N m)]
10^{-13}
10^{-15}
10^{-17}
10^{-19}
10^{-21}
10^{12}
10^{13}
10^{14}
10^{15}
10^{16}
10^{17}
10^{18}
seismic moment M_0 [N m]